

01. Vector Space

Exercise 1. Let $M = \begin{pmatrix} 2 & 5 & 3 \\ 1 & 0 & -7 \end{pmatrix}$ and $N = \begin{pmatrix} 4 & -2 & 5 \\ -5 & 3 & 2 \end{pmatrix}$. Determine the following.

- (a) M_{13}
- (b) $M + N$
- (c) $5M$

Exercise 1.1. The following tables indicate how many uniforms that school A and B ordered in 2009 and 2010, respectively. Write the table for the number of uniforms the school ordered in sum.

	Red	Blue	White		Red	Blue	White
School A	300	200	400	School A	250	300	350
School B	500	200	300	School B	400	250	350
	(2009)				(2010)		

Exercise 1.2. Write the zero vector of $M_{3 \times 4}(F)$.

Exercise 1.3. Consider the following matrices.

$$A = \begin{pmatrix} 2 & -1 \\ 0 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 8 & -3 \\ 0 & 1 \end{pmatrix}, \quad C = \begin{pmatrix} 10 & x \\ 0 & y \end{pmatrix},$$

Find $x, y \in \mathbb{R}$ which makes $A + B = C$.

Exercise 2. Let V be the set of all differentiable real-valued functions defined on $\mathbb{R}$. Show that V is a vector space with the obvious addition and scalar multiplication.

Exercise 2.1. Let V be the set consisting of single element. Define addition and scalar multiplication on V to give the vector space structure. Here, V is called the **zero vector space**.

Exercise 2.2. Let $V = \mathbb{R}^2$. Determine whether V is a vector space with the given addition and multiplication. Here, $(a_1, a_2), (b_1, b_2) \in V$ and $c \in \mathbb{R}$.

- (a) $(a_1, a_2) + (b_1, b_2) = (a_1 + b_1, a_2 b_2)$, $c(a_1, a_2) = (ca_1, a_2)$
- (b) $(a_1, a_2) + (b_1, b_2) = (a_1 + 2b_1, a_2 + 3b_2)$, $c(a_1, a_2) = (ca_1, ca_2)$
- (c) $(a_1, a_2) + (b_1, b_2) = (a_1 + b_1, a_2 + b_2)$, $c(a_1, a_2) = \begin{cases} (0, 0) & \text{if } c = 0 \\ (ca_1, a_2/c) & \text{if } c \neq 0 \end{cases}$

Exercise 2.3. Consider $V = \mathbb{C}^n$ with the usual vector space structure over $\mathbb{C}$. Show that V is also a vector space over $\mathbb{R}$.

Exercise 3. Let V be a vector space over a field F and let $a \in F$, $x, y \in V$.

- (a) Show that if $ax = 0$, then $a = 0$ or $x = 0$.
- (b) Is it true that if $ax = ay$, then $x = y$?

Exercise 3.1. Let V be a vector space. For $a, b \in F$, $x, y \in V$, show that the following identity holds.

$$(a + b)(x + y) = ax + ay + bx + by$$

Exercise 3.2. Let V be a vector space. For $0 \neq a \in F$ and $x \in V$, show that there is a unique $y \in V$ such that $ay + x = 0$.

Exercise 4. Let $A = \begin{pmatrix} -4 & 2 \\ 5 & -1 \end{pmatrix}$. Determine the following.

- (a) A^t
- (b) $\text{tr}(A)$

Exercise 4.1. (a) For $A \in M_{m \times n}(F)$, show that $(A^t)^t = A$.

(b) For $A \in M_{n \times n}(F)$, show that $A + A^t$ is symmetric.

(c) For $A, B \in M_{n \times n}(F)$ and $a, b \in F$, show that $\text{tr}(aA + bB) = a\text{tr}(A) + b\text{tr}(B)$.

(d) For $A \in M_{n \times n}(F)$, show that $\text{tr}(A^t) = \text{tr}(A)$.

Exercise 5. Determine whether the following sets are subspaces of $\mathbb{R}^3$ with the usual vector space structure.

- (a) $W = \{(a_1, a_2, a_3) \in \mathbb{R}^3 : a_1 = 3a_2, a_3 = -a_2\}$
- (b) $W = \{(a_1, a_2, a_3) \in \mathbb{R}^3 : a_1 + 2a_2 - 3a_3 = 1\}$
- (c) $W = \{(a_1, a_2, a_3) \in \mathbb{R}^3 : 5a_1^2 - 3a_2^2 + 6a_3^2 = 0\}$

Exercise 5.1. Determine whether the following sets are subspaces of the given vector space V .

- (a) $W = \{f(x) \in P(F) : f(x) = 0 \text{ or } f(x) \text{ has degree } n\}$, $V = P(F)$
- (b) $W = \{A \in M_{m \times n}(F) : A \text{ is upper triangular}\}$, $V = M_{m \times n}(F)$
- (c) For a nonempty set S and $s_0 \in S$, $W = \{f \in \mathcal{F}(S, F) : f(s_0) = 0\}$, $V = \mathcal{F}(S, F)$
- (d) The set $C^n(\mathbb{R})$ of functions $f : \mathbb{R} \rightarrow \mathbb{R}$ which has a continuous n th derivative, $V = \mathcal{F}(\mathbb{R}, \mathbb{R})$

Exercise 5.2. Let F_1, F_2 be fields. Show that the set of even functions and the set of odd functions are subspaces of $\mathcal{F}(F_1, F_2)$.

Problem 1. Let $S = \{0, 1\}$ and $F = \mathbb{R}$. Consider the following elements in $\mathcal{F}(S, \mathbb{R})$.

$$f(t) = 2t + 1, \quad g(t) = 1 + 4t - 2t^2, \quad h(t) = 5^t + 1$$

Show that $f = g$ and $f + g = h$.

Problem 2. Find the number of elements in $M_{m \times n}(\mathbb{F}_p)$ where p is a prime number.

Problem 3. Let V be the set of positive real numbers. For $x, y \in V$ and $c \in \mathbb{R}$, define $x + y := xy$ and $cx := x^c$ where the right hand sides are given by the usual operations in $\mathbb{R}$. Show that V is a vector space over $\mathbb{R}$.

Problem 4. Let V be a vector space and W_1, W_2 be two subspaces of V . Show that $W_1 \cup W_2$ is a subspace of V if and only if $W_1 \subset W_2$ or $W_2 \subset W_1$.

Problem 5. Let V be a vector space. For nonempty subsets S_1, S_2 of V , we define their **sum** by $S_1 + S_2 = \{x_1 + x_2 : x_1 \in S_1, x_2 \in S_2\}$.

- (1) Let W_1, W_2 be subspaces of V . Show that $W_1 + W_2$ is a subspace of V that contains both W_1 and W_2 . Furthermore, show that if a subspace W of V contains both W_1 and W_2 , then W contains $W_1 + W_2$.
- (2) Let W be a subspace of V . For $v \in V$, $v + W := \{v\} + W$ is called the **coset** of W containing v . Consider the set $S = \{v + W : v \in V\}$ and define the addition and scalar multiplication as follows. (Here, $v_1, v_2 \in W$ and $a \in F$.)

$$(v_1 + W) + (v_2 + W) = (v_1 + v_2) + W, \quad a(v + W) = av + W$$

Show that the operations are well-defined and S becomes a vector space with respect to the operations. The vector space is called **the quotient space** V modulo W and denoted by V/W .

Problem 6. A vector space V is called the **direct sum** of W_1 and W_2 if W_1, W_2 are subspaces of V such that $W_1 \cap W_2 = 0$ and $W_1 + W_2 = V$. In this case, we write $V = W_1 \oplus W_2$.

- (1) Consider the following subsets of $M_{m \times n}(F)$.

$$W_1 = \{A \in M_{m \times n}(F) : A_{ij} = 0 \text{ for all } i > j\}$$

$$W_2 = \{A \in M_{m \times n}(F) : A_{ij} = 0 \text{ for all } i \leq j\}$$

Show that $M_{m \times n}(F) = W_1 \oplus W_2$.

- (2) A matrix A is called **skew symmetric** if $A^t = -A$. Let W_1 be the set of $n \times n$ skew-symmetric matrices over $\mathbb{R}$ and let W_2 be the set of $n \times n$ symmetric matrices over $\mathbb{R}$. Show that $M_{n \times n}(F) = W_1 \oplus W_2$.
- (3) Let W_1, W_2 be subspaces of a vector space V . Show that $V = W_1 \oplus W_2$ if and only if each $v \in V$ can be uniquely written as $x_1 + x_2$ where $x_1 \in W_1$ and $x_2 \in W_2$.

02. Span and Linear Independence

Exercise 1. Determine whether the first vector in a vector space V can be expressed as a linear combination of the other two.

(a) $(-2, 0, 3), (1, 3, 0), (2, 4, -1)$ where $V = \mathbb{R}^3$

(b) $x^3 - 3x + 5, x^3 + 2x^2 - x + 1, x^3 + 3x^2 - 1$ where $V = P_3(\mathbb{R})$

Exercise 1.1. Determine whether the first vector in a vector space V can be expressed as a linear combination of the other vectors.

(a) $\begin{pmatrix} 1 & 2 \\ -3 & 4 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ where $V = M_{2 \times 2}(\mathbb{R})$

(b) $f(t) = t^3 + t, g(t) = t + 1, h(t) = t^2 - 1$ where $V = \mathcal{F}(S, \mathbb{R}), S = \{0, 1\}$

Exercise 2.

(a) Let e_i be the vector in F^n whose i th coordinate is 1 and whose other coordinates are 0. Show that $\{e_1, \dots, e_n\}$ generates F^n .

(b) Show that $P_n(F)$ is generated by $\{1, x, \dots, x^n\}$.

Exercise 2.1. Consider the following matrices.

$$M_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad M_2 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad M_3 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Show that the span of $\{M_1, M_2, M_3\}$ is the set of all symmetric 2×2 matrices.

Exercise 2.2. Let V be a vector space over F .

(a) For $x \in V$, find $\text{span}(\{x\})$ explicitly.

(b) For $x_1, \dots, x_n \in V$, find $\text{span}(\{x_1, \dots, x_n\})$ explicitly.

Exercise 3. Let S be a subset of a vector space V . Show that $\text{span}(S)$ is the intersection of all subspaces of V containing S .

Exercise 3.1. Let W be a subset of a vector space V . Show that W is a subspace of V if and only if $\text{span}(W) = W$

Exercise 3.2. Let S_1, S_2 be subsets of a vector space V . Show the following statements.

(a) If $S_1 \subset S_2$, then $\text{span}(S_1) \subset \text{span}(S_2)$.

(b) If $S_1 \subset S_2$ and $\text{span}(S_1) = V$, then $\text{span}(S_2) = V$.

Exercise 4. Determine whether the following sets are linearly dependent or linearly independent.

- (a) $\{(1, -1, 2), (1, -2, 1), (1, 1, 4)\}$ in $\mathbb{R}^3$
- (b) $\{x^3 + 2x^2, -x^2 + 3x + 1, x^3 - x^2 + 2x - 1\}$ in $P_3(\mathbb{R})$

Exercise 4.1. Determine whether the following sets are linearly dependent or linearly independent.

- (a) $\left\{ \begin{pmatrix} 1 & -3 \\ -2 & 4 \end{pmatrix}, \begin{pmatrix} -2 & 6 \\ 4 & -8 \end{pmatrix} \right\}$ in $M_{2 \times 2}(\mathbb{R})$
- (b) $\left\{ \begin{pmatrix} 1 & 1 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 1 \\ 0 & 1 \end{pmatrix} \right\}$ in $M_{3 \times 2}(\mathbb{R})$

Exercise 4.2. Consider functions $f, g \in \mathcal{F}(\mathbb{R}, \mathbb{R})$ defined by $f(t) = e^{rt}$ and $g(t) = e^{st}$. Show that $\{f, g\}$ are linearly independent if and only if $r \neq s$.

Exercise 4.3.

- (a) Let e_i be the vector in F^n whose i th coordinate is 1 and whose other coordinates are 0. Show that $\{e_1, \dots, e_n\}$ is linearly independent.
- (b) Show that $\{1, x, \dots, x^n\}$ is linearly independent in $P_n(F)$.
- (c) Let E^{ij} be the $m \times n$ matrix whose only nonzero entry is 1 in the i th row and j th column. Show that $\{E^{ij} : 1 \leq i \leq m, 1 \leq j \leq n\}$ is linearly independent and spans $M_{m \times n}(F)$.

Exercise 5.

- (a) Let u and v be distinct vectors of a vector space V . Show that $\{u, v\}$ is linearly dependent if and only if u or v is a multiple of the other.
- (b) Find an example of three linearly dependent vectors in $\mathbb{R}^3$ such that none of the three is a multiple of another.

Exercise 5.1. Let $S = \{u_1, \dots, u_n\}$ be a subset of a vector space V . Prove that S is linearly dependent if and only if $u_1 = 0$ or $u_{k+1} \in \text{span}(\{u_1, \dots, u_k\})$ for some k ($1 \leq k < n$).

Exercise 5.2. Let S be a subset of a vector space V . Show that S is linearly independent if and only if every finite subset of S is linearly independent.

Problem 1. Let S_1, S_2 be subsets of a vector space V . Show the following.

- (a) $\text{span}(S_1 \cup S_2) = \text{span}(S_1) + \text{span}(S_2)$
- (b) $\text{span}(S_1 \cap S_2) \subset \text{span}(S_1) \cap \text{span}(S_2)$

Problem 2. Let $S = \{(1, 1, 0), (1, 0, 1), (0, 1, 1)\}$ be a subset of the vector space F^3 . Prove the following.

- (a) If $F = \mathbb{R}$, then S is linearly independent.
- (b) If $F = \mathbb{F}_2$, then S is linearly dependent.

Problem 3. Let V be a vector space over $\mathbb{R}$.

- (a) For distinct vectors u, v in V , show that $\{u, v\}$ is linearly independent if and only if $\{u + v, u - v\}$ is linearly independent.
- (b) For distinct vectors u, v, w in V , show that $\{u, v, w\}$ is linearly independent if and only if $\{u + v, v + w, w + u\}$ is linearly independent.

Problem 4. Let S_1, S_2 be disjoint linearly independent subsets of V . Show that $S_1 \cup S_2$ is linearly independent if and only if $\text{span}(S_1) \cap \text{span}(S_2) = \{0\}$

Problem 5. Let $\{A_1, \dots, A_k\}$ be a linearly independent subset of $M_{n \times n}(F)$. Prove that $\{A_1^t, \dots, A_k^t\}$ is linearly independent.

Problem 6. Let M be a square upper triangular matrix with nonzero diagonal entries. Prove that the columns of M are linearly independent.

03. Bases and Dimension

Exercise 1. Determine whether the following sets are bases for a vector space V .

- (a) $\{(1, 0, -1), (2, 5, 1), (0, -4, 3)\}$, $V = \mathbb{R}^3$
- (b) $\{(1, 4, -6), (1, 5, 8), (2, 1, 1), (0, 1, 0)\}$, $V = \mathbb{R}^3$
- (c) $\{(1 + 2x + x^2, 3 + x^2, x + x^2)\}$, $V = P_2(\mathbb{R})$
- (d) $\{(x^3 - 2x^2 + 1, 4x^2 - x + 3, 3x - 2)\}$, $V = P_3(\mathbb{R})$

Exercise 1.1. Give three different bases for $M_{2 \times 2}(F)$.

Exercise 1.2. Consider five vectors $u_1 = (2, -3, 1)$, $u_2 = (1, 4, -2)$, $u_3 = (-8, 12, -4)$, $u_4 = (1, 37, -17)$, and $u_5 = (-3, -5, 8)$. Find a subset of $\{u_1, u_2, u_3, u_4, u_5\}$ which is a basis of $\mathbb{R}^3$.

Exercise 1.3. Let u, v, w be distinct vectors of a vector space V and let a, b be nonzero scalars.

- (a) Show that if $\{u, v\}$ is a basis for V , then $\{u + v, au\}$ and $\{au, bv\}$ are bases for V .
- (b) Show that if $\{u, v, w\}$ is a basis for V , then $\{u + v + w, v + w, w\}$ is a basis for V .

Exercise 2. Let $V = \{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1 - 2x_2 + x_3 = 0, 2x_1 - 3x_2 + x_3 = 0\}$.

- (1) Show that V is a subspace of $\mathbb{R}^3$.
- (2) Find a basis of V .
- (3) Find $\dim V$.

Exercise 2.1. Find the dimensions of the given subspaces of $M_{n \times n}(F)$.

- (a) the set of all $n \times n$ matrices having trace equal to zero
- (b) the set of all $n \times n$ upper triangular matrices
- (c) the set of all $n \times n$ skew-symmetric matrices
(Recall that a square matrix A is skew-symmetric if $A^t = -A$.)

Exercise 2.2. Let V be the vector space of all sequences in F . Let W be the set of sequences in F that have only a finite number of nonzero terms.

- (a) Show that W is a subspace of V .
- (b) Find a basis of W .
- (c) Find $\dim W$.

Exercise 2.3. Fix $a \in \mathbb{R}$. Find the dimension of the subspace of $P_n(\mathbb{R})$ defined by

$$\{f \in P_n(\mathbb{R}) : f(a) = 0\}$$

Exercise 3. Let V be a vector space and let W_1 and W_2 be subspaces of V . Find a necessary and sufficient conditions on W_1 and W_2 so that $\dim(W_1 \cap W_2) = \dim W_2$.

Exercise 3.1. Let V be a vector space and take $v_1, v_2, \dots, v_k, v \in V$. Let $W_1 = \text{span}(\{v_1, v_2, \dots, v_k\})$ and $W_2 = \text{span}(\{v_1, v_2, \dots, v_k, v\})$. Find a necessary and sufficient conditions on v so that $\dim W_1 = \dim W_2$

Problem 1. Find all vector spaces V which have only a finite number of generating sets. (Hint. Consider the case F is finite and the case F is infinite separately.)

Problem 2. Let $f(x)$ be a polynomial of degree n in $P_n(\mathbb{R})$. Show that $\{f(x), f'(x), \dots, f^{(n)}(x)\}$ is a basis of $P_n(\mathbb{R})$.

Problem 3.

- (1) Let V be a vector space of dimension n and let S be a generating set of V . Show that there is a subset of S which is a basis of V .
- (2) Prove that a vector space is infinite-dimensional if and only if it contains an infinite linearly independent subset.

Problem 4. Show that if V is a vector space over $\mathbb{C}$ of dimension n , then V is a vector space over $\mathbb{R}$ of dimension $2n$. (Here, n is a nonnegative integer.)

Problem 5. Let V be a vector space and let W_1 and W_2 be finite-dimensional subspaces of V . Show the following.

- (1) $\dim(W_1 + W_2) = \dim W_1 + \dim W_2 - \dim(W_1 \cap W_2)$
- (2) $V = W_1 \oplus W_2$ if and only if $V = W_1 + W_2$ and $\dim V = \dim W_1 + \dim W_2$.

Problem 6. Let V be a finite-dimensional vector space and let W be a subspace of V .

- (1) Show that there exists a subspace W' of V such that $V = W \oplus W'$.
- (2) Show that W' in (1) may not be unique.

Problem 7. Let V be a finite-dimensional vector space and let W be a subspace of V . Show that $\dim(V/W) = \dim V - \dim W$.

04. Existence of a Basis

Exercise 1. Let $\mathcal{F}$ be a family of subsets T of $S = \{1, \dots, 4\}$ such that the sum of elements of T is odd.

- (1) Find all maximal elements in $\mathcal{F}$.
- (2) Find a chain in $\mathcal{F}$ which has the largest number of elements.

Exercise 1.1. Find an example of a family $\mathcal{F}$ of sets such that $\mathcal{F}$ is infinite but every chain in $\mathcal{F}$ is finite.

Exercise 2. Let S_1, S_2 be subsets of a vector space V . Show that if S_1 is linearly independent and S_2 generates V , then there exists a subset S'_2 of S_2 such that $S_1 \cup S'_2$ is a basis of V .

Exercise 2.1.

- (1) Let W be a subspace of a vector space V . Show that any basis for W can be extended to a basis for V .
- (2) Let S_1, S_2 be subsets of a vector space V such that $S_1 \subset S_2$. Show that if S_1 is linearly independent and S_2 generates V , then there is a basis β of V such that $S_1 \subset \beta \subset S_2$.
- (3) Let β be a basis of a vector space V and let S be a linearly independent subset of V . Show that there exists a subset S_1 of β such that $S \cup S_1$ is a basis of V .

Exercise 3. Show that the set of convergent sequences in $\mathbb{R}$ is an infinite-dimensional vector space over $\mathbb{R}$.

Exercise 3.1. Show that $\mathbb{R}$ is an infinite-dimensional vector space over $\mathbb{Q}$.

05. Linear Transformation

Exercise 1. Determine whether the following functions are linear.

- (a) $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$, $T(a_1, a_2, a_3) = (a_1 - a_2, 2a_3)$
- (b) $T : P_2(\mathbb{R}) \rightarrow P_3(\mathbb{R})$, $T(f(x)) = xf(x) + f'(x)$
- (c) $T : M_{n \times n}(F) \rightarrow F$, $T(A) = \text{tr}(A)$
- (d) $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $T(a_1, a_2) = (a_1 + 1, a_2)$
- (e) $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $T(a_1, a_2) = (\sin a_1, 0)$

Exercise 1.1. Suppose that $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is linear. Suppose that $T(1, 0) = (1, 4)$ and $T(1, 1) = (2, 5)$. Find the value of $T(2, 3)$.

Exercise 1.2. Find the linear transform $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$, which corresponds to the reflection about the xy -plane.

Exercise 2. For each linear transform T in Exercise 1, compute the nullity and rank of T and determine whether T is injective or surjective.

Exercise 2.1. Let V be the space of real sequences. Define $T, U : V \rightarrow V$ as follows.

$$T(a_1, a_2, \dots) = (a_2, a_3, \dots), \quad U(a_1, a_2, \dots) = (0, a_1, a_2, \dots)$$

- (1) Show that T, U are linear.
- (2) Show that T is onto but not one-to-one.
- (3) Show that U is one-to-one but not onto.

Exercise 2.2. Find an example of two distinct linear transforms $T, U : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ such that $\text{Ker}(T) = \text{Ker}(U)$ and $\text{Im}(T) = \text{Im}(U)$

Exercise 2.3. Let $T : V \rightarrow W$ be a linear transform.

- (1) Show that $T^{-1}(0) = \text{Ker}(T)$
- (2) Show that if $T(x) = y$, then $T^{-1}(y) = x + \text{Ker}(T)$

Problem 1. Let $T : F^n \rightarrow F$ be a linear transform. Show that there exist $a_1, \dots, a_n \in F$ such that $T(x_1, \dots, x_n) = a_1x_1 + \dots + a_nx_n$ for all $x_1, \dots, x_n \in F$.

Problem 2. Let V be a vector space and W_1, W_2 be subspaces of V such that $V = W_1 \oplus W_2$. The function $T : V \rightarrow V$ defined by $T(x_1 + x_2) = x_1$ for $x_1 \in W_1, x_2 \in W_2$ is called the **projection** of V on W_1 (along W_2).

- (1) Show that $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T(a, b, c) = (a - c, b, 0)$ is the projection on the xy -plane along the line $L = \{(a, 0, a) : a \in \mathbb{R}\}$.
- (2) Find the linear transform $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ where T represents the projection on the y -axis along the line $L = \{(s, s) : s \in \mathbb{R}\}$.

Problem 3. Let $T : V \rightarrow V$ be a linear transform. A subspace W of V is called **T -invariant** if $T(x) \in W$ for all $x \in W$. In this case, $T|_W : W \rightarrow W$ defined by $T|_W(x) = T(x)$ is called the **restriction** of T on W .

- (1) Let $T : V \rightarrow V$ be a linear transform. Show that $0, V, \text{Ker}(T), \text{Im}(T)$ are T -invariant subspaces of V .
- (2) Let $T : V \rightarrow V$ be the projection on W_1 along W_2 . Show that W_1 is T -invariant.
- (3) Let $T : V \rightarrow V$ be a linear transform and let W be a T -invariant subspace of V . Show that $\text{Ker}(T|_W) = \text{Ker}(T) \cap W$ and $\text{Im}(T|_W) = T(W)$.

Problem 4. A function $T : V \rightarrow W$ between vector spaces is called **additive** if $T(x + y) = T(x) + T(y)$ for all $x, y \in V$.

- (1) Show that if V, W are vector spaces over $\mathbb{Q}$ and $T : V \rightarrow W$ is additive, then T is linear.
- (2) Find an additive function that is not linear.

Problem 5. Let V be a finite-dimensional vector space and let $T : V \rightarrow V$ be a linear transform. Show that $V = \text{Ker}(T) + \text{Im}(T)$ if and only if $\text{Ker}(T) \cap \text{Im}(T) = 0$.

06. Linear Transformation and Matrix

Exercise 1. For each linear transform $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and bases β and γ of $\mathbb{R}^n$ and $\mathbb{R}^m$ below, compute $[T]_{\beta}^{\gamma}$.

- (1) $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$, $T(a_1, a_2) = (2a_1 - a_2, 3a_1 + 4a_2, a_1)$, β and γ are standard ordered bases.
- (2) $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$, $T(a_1, a_2) = (a_1 - a_2, a_1, 2a_1 + a_2)$,
 $\beta = \{(1, 2), (2, 3)\}$ and $\gamma = \{(1, 1, 0), (0, 1, 1), (2, 2, 3)\}$.

Exercise 1.1. Let E^{ij} be the matrix whose only nonzero entry is 1 in the i th row and j th column. Consider the following.

$$\alpha = \{E^{11}, E^{12}, E^{21}, E^{22}\}, \quad \beta = \{1, x, x^2\}, \quad \gamma = \{1\}$$

- (1) Define $T : M_{2 \times 2}(F) \rightarrow M_{2 \times 2}(F)$ by $T(A) = A^t$. Compute $[T]_{\alpha}$.
- (2) Define $T : M_{2 \times 2}(F) \rightarrow F$ by $T(A) = \text{tr}(A)$. Compute $[T]_{\alpha}^{\gamma}$.
- (3) Define $T : P_2(\mathbb{R}) \rightarrow M_{2 \times 2}(\mathbb{R})$ by $T(f(x)) = \begin{pmatrix} f'(0) & 2f(1) \\ 0 & f''(3) \end{pmatrix}$. Compute $[T]_{\beta}^{\alpha}$.

Exercise 1.2. Let $V = \mathbb{C}$ regarded as a vector space over $\mathbb{R}$ and let $\beta = \{1, i\}$ be a basis for V . Define $T : V \rightarrow V$ by $T(z) = \bar{z}$. Compute $[T]_{\beta}$.

Exercise 1.3. Let V be a vector space and let $\beta = \{v_1, \dots, v_n\}$ be a basis for V . Define a linear transform $T : V \rightarrow V$ by $T(v_i) = v_i + v_{i-1}$ for $i = 1, \dots, n$ where $v_0 = 0$. Compute $[T]_{\beta}$.

Exercise 2. Let $V = P(\mathbb{R})$ and define $T_j : V \rightarrow V$ by $T_j(f(x)) = f^{(j)}(x)$ for each $j \geq 1$. Show that $\{T_j : j \geq 1\}$ is a linearly independent subset of $\mathcal{L}(V)$.

Exercise 2.1. Let $T, U : V \rightarrow W$ be nonzero linear transforms. Show that if $\text{Im}T \cap \text{Im}U = 0$, then $\{T, U\}$ is a linearly independent subset of $\mathcal{L}(V, W)$.

Exercise 3. Let

$$A = \begin{pmatrix} 1 & 3 \\ 2 & -1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 & -3 \\ 4 & 1 & 2 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 1 & 4 \\ -1 & -2 & 0 \end{pmatrix}$$

- (1) Compute $A(2B + 3C)$.
- (2) Compute $C^t A$.

Exercise 3.1. The matrices A, B, C are of size 5×6 , 5×6 , 6×3 , respectively. Determine whether the given multiplication is possible. If possible, compute its size.

- (1) $B + AB$ (2) $A^t B$ (3) $B^t AC$

Exercise 3.2. (1) Find matrices A, B such that $AB = O$ but $A \neq O$ and $B \neq O$.

- (2) Find a nonzero matrix A such that $A^2 = O$.

Exercise 3.3. Let A, B be $n \times n$ matrices.

- (1) Show that $\text{tr}(AB) = \text{tr}(BA)$.
 (2) Show that $\text{tr}(A^t) = \text{tr}(A)$.
 (3) Show that there are no matrices A, B such that $AB - BA = I$.

Exercise 4. Define $T : P_2(\mathbb{R}) \rightarrow P_2(\mathbb{R})$ and $U : P_2(\mathbb{R}) \rightarrow \mathbb{R}^3$ by

$$T(f(x)) = f'(x)(3 + x) + 2f(x), \quad U(a + bx + cx^2) = (a + b, c, a - b).$$

Let β, γ be the standard ordered bases of $P_2(\mathbb{R}), \mathbb{R}^3$, respectively.

- (1) Compute $[U]_\beta^\gamma, [T]_\beta, [UT]_\beta^\gamma$. Find a relation among these.
 (2) Let $h(x) = 3 - 2x + x^2$. Compute $[U]_\beta^\gamma, [h(x)]_\beta, [U(h(x))]_\gamma$. Find a relation among these.

Exercise 4.1. Find a matrix A corresponding to each linear transform with respect to the standard bases. Describe the linear transform corresponding to each A^2 .

- (1) clockwise rotation by θ around the origin $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$
 (2) reflection about the y -axis $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$
 (3) projection on the y -axis $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$

Exercise 4.2. Let $\beta = \{(1, 0), (1, 1)\}$ and $\gamma = \{(0, 1), (2, 1)\}$. Find a linear transform $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ such that $[T]_\beta^\gamma = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.

Exercise 5. Determine whether the given linear transformation T is invertible.

- (1) $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3, T(a_1, a_2) = (a_1 - 2a_2, a_2, 3a_1 + 4a_2)$.
 (2) $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3, T(a_1, a_2, a_3) = (3a_1 - 2a_3, a_2, 3a_1 + 4a_2)$.

Exercise 5.1. Which of the following pairs of vector spaces are isomorphic?

- (1) F^3 and $P_3(F)$
 (2) $V = \{A \in M_{2 \times 2}(\mathbb{R}) : \text{tr}(A) = 0\}$ and $\mathbb{R}^3$

Exercise 5.2. Construct an isomorphism from $V = \left\{ \begin{pmatrix} a & a+b \\ 0 & c \end{pmatrix} : a, b, c \in F \right\}$ to F^3 .

Exercise 6. Let $A, B \in M_{n \times n}(F)$ be invertible. Prove the following statements.

- (1) $(A^{-1})^{-1} = A$
- (2) $(AB)^{-1} = B^{-1}A^{-1}$
- (3) $(A^t)^{-1} = (A^{-1})^t$
- (4) $(A^{-1})^n = (A^n)^{-1}$ for $n \in \mathbb{Z}^{\geq 0}$
- (5) $(cA)^{-1} = c^{-1}A^{-1}$ for $c \in F - \{0\}$

Exercise 6.1. Prove the following.

- (1) If A is invertible and $AB = O$, then $B = O$.
- (2) If $AB = O$ for some nonzero matrix B , then A is not invertible.

Exercise 6.2. Let B be an $n \times n$ invertible matrix. Show that $\Phi : M_{n \times n}(F) \rightarrow M_{n \times n}(F)$ defined by $\Phi(A) = B^{-1}AB$ is an isomorphism.

Problem 1. Let $T : V \rightarrow V$ be a linear transform and let W be a T -invariant subspace of V . Let $\dim V = n$ and $\dim W = k$. Find a basis β for V such that $[T]_\beta$ is of the form

$$\begin{pmatrix} A & B \\ O & C \end{pmatrix}$$

where A is a square matrix of size k and O is the zero matrix of size $(n - k) \times k$.

Problem 2. Let $T : V \rightarrow V$ be the projection on W along W' where $V = W \oplus W'$. Find a basis β for V such that $[T]_\beta$ is a diagonal matrix.

Problem 3. Let $T : V \rightarrow V$ be a linear transform where V is a finite-dimensional vector space.

(1) Show that if $\text{rank}(T) = \text{rank}(T^2)$, then $V = \ker(T) \oplus \text{Im}(T)$

(2) Show that $V = \ker(T^k) \oplus \text{Im}(T^k)$ for some positive integer k .

Problem 4. Let $T : V \rightarrow V$ be a linear transform. Show that $T = T^2$ if and only if T is a projection.

Problem 5. Let A, B be $n \times n$ matrices. Prove the following statements.

(1) If AB is invertible, then A and B are invertible.

(2) If $AB = I$, then $BA = I$.

Problem 6. Find an isomorphism between the space of all sequences in F that have only a finite number of nonzero terms and $P_n(F)$.

Problem 7. (First Isomorphism Theorem)

Let $T : V \rightarrow W$ be a linear transform. Define

$$\bar{T} : V/\text{Ker}T \rightarrow \text{Im}T, \quad \bar{T}(v + \text{Ker}T) = T(v).$$

(1) Show that $\bar{T}$ is a well-defined isomorphism.

(2) Show that $T = \bar{T} \circ P$ where $P : V \rightarrow V/\text{Ker}T$ is defined by $P(v) = v + \text{Ker}T$.

Problem 8. Let V be a vector space and let S be its basis. Let $C(S, F)$ be the space of functions $f : S \rightarrow F$ such that $f(s) = 0$ for all but a finite number of $s \in S$. Show that $\Psi : C(S, F) \rightarrow V$ defined by

$$\Psi(f) = \sum_{s \in S} f(s)s$$

is an isomorphism.

07. Change of Coordinates

Exercise 1. Let T be the linear operator on $\mathbb{R}^2$ defined by

$$T(a, b) = (2a + b, a - 3b).$$

Let β be the standard ordered basis for $\mathbb{R}^2$ and let $\beta' = \{(1, 1), (1, 2)\}$.

- (1) Find the matrix that changes β -coordinates to β' -coordinates.
- (2) Compute $[T]_{\beta'}$.

Exercise 1.1. For each of the following pairs of ordered bases β, β' for the given vector space V , find the matrix that changes β' -coordinates to β -coordinates.

- (1) $V = \mathbb{R}^2$, $\beta = \{(-1, 3), (2, -1)\}$, $\beta' = \{(0, 10), (5, 0)\}$
- (2) $V = P_2(\mathbb{R})$, $\beta = \{x^2, x, 1\}$, $\beta' = \{a_2x^2 + a_1x + a_0, b_2x^2 + b_1x + b_0, c_2x^2 + c_1x + c_0\}$

Exercise 1.2. Let $A = \begin{pmatrix} 1 & 3 \\ 1 & 1 \end{pmatrix}$ and $\beta = \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right\}$. Find an invertible matrix Q such that $[L_A]_{\beta} = Q^{-1}AQ$.

Exercise 1.3. Find an expression for the reflection of $\mathbb{R}^2$ about the line $L : y = mx$ ($m \neq 0$)

Exercise 2. Suppose that A, B are similar $n \times n$ matrices. Prove the following.

- (1) If A, B are invertible, then A^{-1}, B^{-1} are similar.
- (2) A^t, B^t are similar.

Exercise 2.1. Suppose that A, B are similar $n \times n$ matrices.

- (1) Show that $\text{tr}(A) = \text{tr}(B)$.
- (2) Describe how to define the trace of a linear transform and show that it is well-defined.

08. Dual Space

Exercise 1. Let $V = \mathbb{R}^3$.

- (1) Let $\beta = \{(1, 0, 1), (1, 2, 1), (0, 0, 1)\}$. Find the dual basis β^* of V^* .
- (2) Define $f_1, f_2, f_3 \in V^*$ as follows.

$$f_1(x, y, z) = x - 2y, \quad f_2(x, y, z) = x + y + z, \quad f_3(x, y, z) = y - 3z.$$

Prove that $\{f_1, f_2, f_3\}$ is a basis for V^* and find the basis of V whose dual basis is $\{f_1, f_2, f_3\}$.

Exercise 1.1. Determine whether f is a linear functional on a vector space V .

- (1) $V = P(\mathbb{R})$, $f(p(x)) = 2p'(0) + p''(1)$.
- (2) $V = \mathbb{R}^3$, $f(x, y, z) = x^2 + y^2 + z^2$.
- (3) $V = \mathbb{R}^2$, $f(x, y) = (2x, 4y)$.

Exercise 1.2. Let $\{u, v\}$ be a linearly independent set in $\mathbb{R}^3$. Find a vector in $(\mathbb{R}^3)^*$ whose null space is the plane $\{su + tv : s, t \in \mathbb{R}\}$.

Exercise 1.3. Let $V = P_n(F)$ and let $c_0, \dots, c_n$ be distinct scalars in F . Define $f_i \in V^*$ by $f_i(p(x)) = p(c_i)$. ($0 \leq i \leq n$). Show that $\{f_0, \dots, f_n\}$ is a basis for V^* .

Exercise 2. Define $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by $T(x, y) = (3x + 2y, x)$ and $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ by $f(x, y) = 2x + y$.

- (1) Compute $T^t(f)$.
- (2) Compute $[T]_\beta$ and $[T^t]_{\beta^*}$ where β is the standard basis for $\mathbb{R}^2$. Find a relation between them.

Exercise 2.1. Let V be a finite dimensional vector space and let β be its basis. Consider $\psi : V \rightarrow V^{**}$ defined by $\psi(x)(f) = f(x)$. Show that $\psi(\beta) = \beta^{**}$.

Problem 1. Let V be a finite-dimensional vector space over F . For a subset S of V , the **annihilator** of S is defined by

$$S^\perp = \{f \in V^* : f(x) = 0 \text{ for all } x \in S\}.$$

- (1) Prove that $S^\perp$ is a subspace of V^* .
- (2) Prove that $(S^\perp)^\perp$ is naturally identified with $\text{span}(S)$.
- (3) For subspaces W_1, W_2 , prove that $W_1 = W_2$ if and only if $W_1^\perp = W_2^\perp$.
- (4) For subspaces W_1, W_2 , prove that $(W_1 + W_2)^\perp = W_1^\perp \cap W_2^\perp$.
- (5) For a subspace W of V , prove that $\dim(W) + \dim(W^\perp) = \dim(V)$.

Problem 2. Let $T : V \rightarrow W$ be a linear transform where V, W are finite-dimensional vector spaces over F . Prove the following.

- (1) $\text{Ker}(T^t) = (\text{Im}(T))^\perp$
- (2) $\text{rank}(T) = \text{rank}(T^t)$

Problem 3. Let $T : V \rightarrow W$ be a linear transform where V, W are nonzero finite-dimensional vector spaces over F . Prove the following.

- (1) T is surjective if and only if T^t is injective.
- (2) T is injective if and only if T^t is surjective.

Problem 4. Let $T : V \rightarrow V$ be a linear transform where $\dim V < \infty$ and let W be a subspace of V . Prove that W is T -invariant if and only if $W^\perp$ is T^t -invariant.

09. Elementary Matrix Operation

Exercise 1. Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 0 & 1 \\ 1 & -1 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 0 & 3 \\ 1 & -2 & 1 \\ 1 & -3 & 1 \end{pmatrix}$.

(1) Find an elementary operation that transforms A into B .

(2) Perform elementary row operations to transform B into I_3 .

Exercise 1.1. Find the inverse of the following matrices.

$$(a) \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \quad (b) \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (c) \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix}$$

Exercise 1.2. Prove that there is a sequence of elementary row operations of type 1 and 3 that transforms an $m \times n$ matrix A into an upper triangular matrix.

Exercise 1.3. Prove that an elementary row operation of type 1 can be obtained by a succession of elementary row operations of type 2 and 3.

Exercise 2. Find the rank of the following matrices.

$$(a) \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix} \quad (b) \begin{pmatrix} 1 & 0 & 2 \\ 1 & 1 & 4 \end{pmatrix} \quad (c) \begin{pmatrix} 1 & 2 & 3 & 1 & 1 \\ 1 & 4 & 0 & 1 & 2 \\ 0 & 2 & -3 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Exercise 2.1. Decompose $A = \begin{pmatrix} 2 & 1 \\ -1 & 2 \\ 2 & 1 \end{pmatrix}$ into the form $A = PDQ$ where P, Q are invertible matrices and $D = \begin{pmatrix} I & O \\ O & O \end{pmatrix}$

Exercise 2.2. Let A be an $m \times n$ matrix. Prove that if c is a nonzero scalar, then $\text{rank}(cA) = \text{rank}(A)$.

Exercise 3. Find the inverse if it exists.

$$(a) \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix} \quad (b) \begin{pmatrix} 1 & 2 & 1 \\ 1 & 3 & 4 \\ 2 & 3 & -1 \end{pmatrix} \quad (c) \begin{pmatrix} 1 & 2 & 1 & 0 \\ 2 & 5 & 5 & 1 \\ -2 & -3 & 0 & 3 \\ 3 & 4 & -2 & -3 \end{pmatrix}$$

Exercise 3.1. Find the inverse of the given linear transform if it exists.

(1) $T : P_2(\mathbb{R}) \rightarrow P_2(\mathbb{R}), T(f(x)) = f''(x) + 2f'(x) - f(x)$

(2) $T : M_{2 \times 2}(\mathbb{R}) \rightarrow \mathbb{R}^4, T(A) = (\text{tr}(A), \text{tr}(A^t), \text{tr}(EA), \text{tr}(AE))$ where

$$E = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Exercise 3.2. Express $\begin{pmatrix} 1 & 2 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 2 \end{pmatrix}$ as a product of elementary matrices.

Exercise 4. Prove the following.

- (1) If A is a 3×1 matrix of rank 1 and B is a 1×3 matrix of rank 1, then AB has rank at most 1.
- (2) If C is a 3×3 matrix of rank 1, then there exist a 3×1 matrix A of rank 1 and 1×3 matrix B of rank 1 such that $C = AB$.

Exercise 4.1. Let A be an $m \times n$ matrix and let B be an $n \times p$ matrix. Show that AB can be written as a sum of n matrices of rank at most one.

Exercise 4.2. Let A be an $m \times n$ matrix of rank m and let B be an $n \times p$ matrix of rank n . Find the rank of AB .

Problem 1. In this problem, say that two matrices P and Q are **equivalent** if a matrix Q can be obtained from a matrix P by a succession of elementary row operations. Show that the relation is indeed an equivalence relation. In other words, prove the following three statements.

- P and P are always equivalent.
- If P and Q are equivalent, then Q and P are equivalent.
- If P and Q are equivalent and Q and S are equivalent, then P and S are equivalent.

Problem 2. Show that if a matrix Q is obtained from P by a succession of elementary row operations, then Q^t is obtained from P^t by a succession of elementary column operations.

Problem 3. For $m \times n$ matrices A and B , show that $\text{rank}(A + B) \leq \text{rank}(A) + \text{rank}(B)$

Problem 4. Prove the following.

- (1) If A is an $m \times n$ matrix of rank m , then there is an $n \times m$ matrix B such that $AB = I_m$.
- (2) If B is an $n \times m$ matrix of rank m , then there is an $m \times n$ matrix A such that $AB = I_m$.

Problem 5. Let

$$A = \begin{pmatrix} 1 & 0 & -1 & 2 & 1 \\ -1 & 1 & 3 & -1 & 0 \\ -2 & 1 & 4 & -1 & 3 \\ 3 & -1 & -5 & 1 & -6 \end{pmatrix}$$

Find the largest rank of a 5×5 matrix B such that $AB = O$.

10. System of Linear Equations

Exercise 1. Consider the following system of linear equations.

$$\begin{aligned}x_1 + x_2 - x_3 &= 0 \\4x_1 + x_2 - 2x_3 &= 0\end{aligned}$$

- (1) Express the system as a single matrix equation.
- (2) Find all solutions to the system.

Exercise 1.1. Consider the following system of linear equations.

$$\begin{aligned}x_1 + 2x_2 - x_3 &= 5 \\x_1 + x_2 + x_3 &= 1 \\2x_1 - 2x_2 + x_3 &= 4\end{aligned}$$

- (1) Express the system as a single matrix equation.
- (2) Solve the system using the inverse of the coefficient matrix.

Exercise 1.2. Define $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ by $T(a, b, c) = (a + b, 2a - c)$. Determine $T^{-1}(1, 11)$.

Exercise 1.3.

- (1) Prove that the system of linear equations $Ax = b$ has a solution if and only if $b \in \text{Im}(L_A)$.
- (2) Define $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ by $T(a, b, c) = (a + b, b - 2c, a + 2c)$. Determine whether $(1, 3, -2) \in \text{Im}T$.

Exercise 2. Use Gaussian elimination to solve the following system of linear equations.

$$\begin{aligned}3x_1 - x_2 + x_3 - x_4 + 2x_5 &= 5 \\x_1 - x_2 - x_3 - 2x_4 - x_5 &= 2 \\5x_1 - 2x_2 + x_3 - 3x_4 + 3x_5 &= 10 \\2x_1 - x_2 - 2x_4 + x_5 &= 5\end{aligned}$$

Exercise 2.1.

- (1) Suppose that the augmented matrix $(A|b)$ is a reduced row echelon form. Show that $Ax = b$ is consistent if and only if $(A|b)$ contains a row in which the only nonzero entry lies in the last column.
- (2) Determine whether the system below is consistent.

$$\begin{aligned}x_1 + 2x_2 - x_3 + x_4 &= 2 \\2x_1 + x_2 + x_3 - x_4 &= 3 \\x_1 + 2x_2 - 3x_3 + 2x_4 &= 2\end{aligned}$$

Exercise 3. The reduced row echelon form of a matrix A is

$$\begin{pmatrix} 1 & 0 & 2 & 0 & -2 \\ 0 & 1 & -5 & 0 & -3 \\ 0 & 0 & 0 & 1 & 6 \end{pmatrix}.$$

(1) Find A if the first, second, and fourth columns of A are

$$\begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix}.$$

(2) Find A if the vectors in (1) are the first, second, and fifth columns of A .

Exercise 3.1. Let W be the space of 2×2 symmetric matrices over $\mathbb{R}$. Consider the set

$$S = \left\{ \begin{pmatrix} 0 & -1 \\ -1 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}, \begin{pmatrix} 2 & 1 \\ 1 & 9 \end{pmatrix}, \begin{pmatrix} 1 & -2 \\ -2 & 4 \end{pmatrix}, \begin{pmatrix} -1 & 2 \\ 2 & -1 \end{pmatrix} \right\}.$$

Find a subset of S that is a basis for W .

Exercise 3.2. Let V be the space of all solutions to the system of linear equations

$$\begin{aligned} x_1 - x_2 + 2x_4 - 3x_5 + x_6 &= 0 \\ 2x_1 - x_2 - x_3 + 3x_4 - 4x_5 + 4x_6 &= 0 \end{aligned}$$

Extend $S = \{(0, -1, 0, 1, 1, 0), (1, 0, 1, 1, 1, 0)\}$ to a basis for V .

11. Definition of Determinants

Exercise 1. Compute the determinants of the following matrices.

$$(1) \begin{pmatrix} 6 & -3 \\ 2 & 4 \end{pmatrix}$$

$$(2) \begin{pmatrix} 5 - 2i & 6 + 4i \\ -3 + i & 7i \end{pmatrix}$$

Exercise 1.1. Compute the area of the parallelogram determined by $u = (3, -2)$ and $v = (2, 5)$.

Exercise 2. Let $A \in M_{2 \times 2}(F)$. Prove the following.

$$(1) \det(A^t) = \det(A).$$

$$(2) \text{ If } B \text{ is a matrix obtained by interchanging the rows of } A, \text{ then } \det(B) = -\det(A).$$

Exercise 2.1. Show that $\det(AB) = \det(A)\det(B)$ for $A, B \in M_{2 \times 2}(F)$.

Exercise 3. Compute the determinants of the following matrices.

$$(1) \begin{pmatrix} 0 & 1 & 2 \\ -1 & 0 & -3 \\ 2 & 3 & 0 \end{pmatrix}$$

$$(2) \begin{pmatrix} 0 & 1 + i & 2 \\ -2i & 0 & 1 - i \\ 3 & 4i & 0 \end{pmatrix}$$

$$(3) \begin{pmatrix} 1 & -1 & 2 & -1 \\ -3 & 4 & 1 & -1 \\ 2 & -5 & -3 & 8 \\ -2 & 6 & -4 & 1 \end{pmatrix}$$

Exercise 3.1. For each equation, find the value of k .

$$(1) \det \begin{pmatrix} 2a_1 & 2a_2 & 2a_3 \\ 3b_1 + 5c_1 & 3b_2 + 5c_2 & 3b_3 + 5c_3 \\ 7c_1 & 7c_2 & 7c_3 \end{pmatrix} = k \det \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix}$$

$$(2) \det \begin{pmatrix} b_1 + c_1 & b_2 + c_2 & b_3 + c_3 \\ a_1 + c_1 & a_2 + c_2 & a_3 + c_3 \\ a_1 + b_1 & a_2 + b_2 & a_3 + b_3 \end{pmatrix} = k \det \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix}$$

Exercise 4. Let $A \in M_{n \times n}(F)$ and $k \in F$. Prove that $\det(kA) = k^n \det(A)$.

Exercise 4.1. Let the rows of $A \in M_{n \times n}(F)$ be $a_1, \dots, a_n$ and consider the matrix B whose rows are $a_n, \dots, a_1$. Express $\det(B)$ in terms of $\det(A)$.

12. Properties of Determinants

Exercise 1. Let A be an $n \times n$ triangular matrix.

- (1) Describe $\det(A)$ in terms of the entries of A .
- (2) Show that A is invertible if and only if its diagonal entries are nonzero.

Exercise 1.1. Let $A \in M_{n \times n}(F)$.

- (1) A matrix A is called **nilpotent** if $A^k = O$ for some integer $k \geq 1$. Show that a nilpotent matrix has the zero determinant.
- (2) A matrix A is called **skew-symmetric** if $A^t = -A$. Show that if $F = \mathbb{C}$ and n is odd, then a skew-symmetric matrix has the zero determinant.
- (3) Let $F = \mathbb{R}$. A matrix A is called **orthogonal** if $AA^t = I$. Show that if A is orthogonal, then $\det(A) = \pm 1$.
- (4) Let $F = \mathbb{C}$. A matrix A is called **unitary** if $AA^* = I$ where $(A^*)_{ij}$ is the complex conjugate of A_{ji} . Show that if A is unitary, then $|\det(A)| = 1$.

Exercise 1.2. Prove that if $A, B \in M_{n \times n}(F)$ are similar, then $\det(A) = \det(B)$.

Exercise 1.3. Suppose that $A, B \in M_{n \times n}(\mathbb{R})$ satisfy $AB = -BA$. Show that if n is odd, then A or B is not invertible.

Exercise 2. Use Cramer's rule to solve the following system of linear equations.

$$\begin{aligned} 2x_1 + x_2 - 3x_3 &= 5 \\ x_1 - 2x_2 + x_3 &= 10 \\ 3x_1 + 4x_2 - 2x_3 &= 0 \end{aligned}$$

Exercise 2.1. Let $A \in M_{n \times n}(F)$ and consider the matrix C where C_{ji} is the cofactor of the row i , column j entry of A . Prove that $AC = (\det A)I$. We call C the **classical adjoint** of A and denote it by $\text{adj}(A)$.

(Hint. Use Cramer's rule for each $Ax = e_j$.)

Exercise 3. Determine whether the following functions $\delta : M_{3 \times 3}(F) \rightarrow F$ is 3-linear.

- (1) $\delta(A) = 1$
- (2) $\delta(A) = A_{11}A_{21}A_{32}$
- (3) $\delta(A) = A_{11}A_{22}A_{33} - A_{11}A_{21}A_{32}$

Exercise 3.1. Find all 2-linear functions $\delta : M_{2 \times 2}(F) \rightarrow F$.

Exercise 3.2. Prove that the set of all n -linear functions over a field F forms a vector space over F .

Problem 1. Suppose that $M \in M_{n \times n}(F)$ can be written in the form

$$M = \begin{pmatrix} A & B \\ O & C \end{pmatrix}$$

where A, C are square matrices. Show that $\det(M) = \det(A)\det(C)$.

Problem 2. A matrix

$$M = \begin{pmatrix} 1 & a_0 & a_0^2 & \cdots & a_0^n \\ 1 & a_1 & a_1^2 & \cdots & a_1^n \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & a_n & a_n^2 & \cdots & a_n^n \end{pmatrix}$$

is called a **Vandermonde matrix**. Prove that

$$\det(M) = \prod_{0 \leq i < j \leq n} (c_j - c_i).$$

Problem 3. Let $y_0(t), y_1(t), \dots, y_n(t) \in C^\infty(\mathbb{R})$. The determinant $W(y_0, y_1, \dots, y_n)(t)$ of a matrix

$$\begin{pmatrix} y_0(t) & y_1(t) & \cdots & y_n(t) \\ y'_0(t) & y'_1(t) & \cdots & y'_n(t) \\ \vdots & \vdots & & \vdots \\ y_0^{(n)}(t) & y_1^{(n)}(t) & \cdots & y_n^{(n)}(t) \end{pmatrix}$$

is called the **Wronskian** of $y_0, \dots, y_n$. Show that $y_0, \dots, y_n$ are linearly independent if $W(y_0, y_1, \dots, y_n)(t) \neq 0$. (The converse also holds.)

Problem 4. Let

$$A = \begin{pmatrix} 0 & 0 & 0 & \cdots & 0 & a_0 \\ -1 & 0 & 0 & \cdots & 0 & a_1 \\ 0 & -1 & 0 & \cdots & 0 & a_2 \\ \vdots & \vdots & \vdots & & \vdots & \\ 0 & 0 & 0 & \cdots & -1 & a_{n-1} \end{pmatrix}$$

Compute $\det(A + tI)$.

Problem 5. Let $A \in M_{n \times n}(F)$. Prove the following.

- (1) $\det(\operatorname{adj}(A)) = (\det A)^{n-1}$.
- (2) $\operatorname{adj}(A^t) = (\operatorname{adj}(A))^t$.
- (3) If A is upper triangular, then $\operatorname{adj}(A)$ is upper triangular.

Problem 6. A **submatrix** of a matrix A is obtained by deleting rows and columns of A . Show that $\operatorname{rank}(A)$ is the largest integer k such that some square submatrix of size k has a nonzero determinant.

Problem 7. Let $\delta : M_{n \times n}(F) \rightarrow F$ be an n -linear function. Suppose that if B is a matrix obtained from $A \in M_{n \times n}(F)$ by interchanging two rows of A , then $\delta(B) = -\delta(A)$. Is it true that δ is alternating?